

Shahzeb Jadoon

AI/ML Engineer

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📍 Rochester, NY

SUMMARY

Edge-AI and ML engineer specializing in hardware-software co-design - deploying deep models on constrained and neuromorphic silicon. Hands-on with INT8 quantization-aware training, custom CUDA C++ kernels, and creating neuromorphic models for real-time autonomous inference at the edge. Strong foundation in computer vision, high-performance computing, and real-time embedded systems, plus production LLM/agent systems.

TOOLS & SKILLS

Edge AI & Hardware-Software Co-Design: INT8/FP16 Quantization, Quantization-Aware Training (QAT), Hardware-Aware Model Adaptation, BrainChip Akida, NVIDIA Jetson, Asymmetric Pruning, ONNX, Edge Latency/Memory Profiling

Systems Programming & High-Performance Computing: C/C++, CUDA, cuBLAS, NVIDIA Nsight Compute, Roofline/CGMA Analysis, FreeRTOS, STM32, MATLAB, Linux/Ubuntu

Machine Learning & Spatial Perception: Python, PyTorch, TensorFlow, Semantic/Instance Segmentation, Object Detection (YOLOv8), Sequence Modeling (RWKV), Depth Estimation, Time-Series Sensor Modeling (EMG, IMU), Spiking Neural Networks (SNNs)

Autonomous Systems & Robotics: CARLA, Multi-Sensor Fusion, HDF5 Data Pipelines, Imitation Learning, Reinforcement Learning, Classical Control (Stanley/PID), Hardware-in-the-Loop (HIL/Quanser), ROS2

LLM Applications & Backend: FastAPI, PostgreSQL, Docker, LiteLLM, Client-Side RAG (BGE Embeddings, In-Browser Retrieval), Cloudflare Zero Trust (RBAC), GitHub Actions (CI/CD)

WORK EXPERIENCE

Machine Learning Engineer Intern

May 2025 - September 2025

BrainChip

Remote (CA)

- Mapped high-resolution vision models onto BrainChip Akida 2.0 custom NPU via end-to-end PyTorch→Akida pipeline, reaching ~82% of the MobileNet baseline accuracy at 7.5% of the parameters (4.6M vs 61.3M).
- Reconfigured models to the accelerator's supported operator set through operator replacement, batch-norm folding, and ONNX graph rewriting (onnxscript); applied INT8 quantization-aware training to recover accuracy.
- Devised a halo-tiling (overlap-tile) inference strategy to run full-resolution segmentation within on-chip memory limits; trained multi-GPU (RTX A6000) on Cityscapes (~0.55 mIoU) and Portrait128 (~0.97 IoU).

Teaching Assistant - Real-Time & Embedded Systems

August 2025 - May 2026

Rochester Institute of Technology

Rochester, NY

- Directed bare-metal C programming and FreeRTOS labs - PWM, DMA, and USART driver implementation.

Teaching Assistant - Introduction to Generative AI

February 2025 - May 2025

Rochester Institute of Technology

Rochester, NY

- Designed curriculum & hands-on PyTorch coding labs, encompassing CNNs, GANs, LLMs, and diffusion models.

ESOP Administrator

July 2022 - June 2023

Palmer Group

Des Moines, IA

- Automated data processing pipelines alongside global administration teams to support a \$5.8B, 75-corporation retirement portfolio, driving a 95% throughput efficiency gain across multi-million user records.

TECHNICAL PROJECTS

Client-Side RAG Portfolio

Personal Website

June 2026

Rochester, NY

- Engineered a serverless, on-device Retrieval-Augmented Generation (RAG) assistant for a personal portfolio utilizing WebLLM and in-browser cosine similarity retrieval.
- Compiled static build-time embedding indexes - zero-latency semantic search without cloud compute overhead.

Production Multi-Agent LLM Platform

Rochester Institute of Technology (RIT)

October 2025 - May 2026

Rochester, NY

- Deployed a secure multi-agent research platform decoupling LLM cognitive processing from physical endpoints via async FastAPI, PostgreSQL 16, and LiteLLM dynamic routing (GPT/Gemini/Claude).
- Hardened the self-hosted Linux perimeter via Cloudflare Zero Trust (identity-aware proxying) and custom HTTP authentication middleware to enforce strict Role-Based Access Control (RBAC).
- Prototyped cyclical agentic workflows via a LangGraph and Model Context Protocol (MCP) scaffold on branch.

Neuromorphic Vision-RWKV - Edge AI Controller

Rochester Institute of Technology (RIT)

January 2026 - May 2026

Rochester, NY

- Benchmarked a neuromorphic-aligned linear attention controller via a pure-PyTorch RWKV-6 temporal encoder, achieving 6.88 ms median inference latency on RTX GPU during offline branch-based benchmarking.
- Architected massively parallel imitation-learning data pipeline on dev branch via CARLA simulation, processing 3.6 TB corpus (3.6M frames) into versioned HDF5 schemas utilizing `posix_fadvise` page-cache eviction.
- Implemented robust classical control fallbacks on dev branch for data collection via Stanley lateral controllers, pure-pursuit following, and dynamic speed planning utilizing NetworkX graph route densification.

VGG-16 Inference Optimization (CUDA)

Rochester Institute of Technology (RIT)

March 2025 - May 2025

Rochester, NY

- Accelerated execution throughput to ~41.5 GFLOPS at batch=16 by coding custom CUDA C++ kernels (shared-memory tiling, memory coalescing, cuBLAS) for convolutional and fully-connected layers.
- Isolated memory-bound bottlenecks utilizing NVIDIA Nsight profiling and roofline/CGMA resource analysis.

Autonomous Driving, CARLA Simulator

Rochester Institute of Technology (RIT)

September 2024 - December 2024

Rochester, NY

- Orchestrated real-time navigation fusing RGB/depth sensors with YOLOv8 and Ultrafast lane detection.

RESEARCH EXPERIENCE

NSF AWARE-AI - Graduate Research Trainee | Rochester Institute of Technology, NY April 2025 - May 2026

- Formulated a semi-supervised 3D monocular depth estimation network in PyTorch as a lightweight, low-cost replacement for LiDAR arrays on mobile robotics platforms.

Resource-Efficient Multimodal Models | Rochester Institute of Technology, NY February 2025 - May 2025

- Investigated model compression techniques for edge-based vision-language model deployment - structured pruning (LLM-Pruner, LoRAPrune) and post-training quantization (GPTQ, AWQ, SpQR).

EDUCATION

Rochester Institute of Technology - Computer Engineering, M.S. (GPA 3.70)

May 2026

Relevant Courses: *High Performance Architectures, Robot Perception, Quantum Machine Learning, Real-time & Embedded Systems, Principles of Robotics, & Self-Adaptive Software w/ RL.*

Wartburg College, Iowa - Computer Science & Actuarial Science, B.A.

May 2020